COURSE BLY 121 GENERAL BIOLOGY II

TOPIC: ECOLOGICAL ADAPTATION OF THE ANIMAL KINGDOM

BY

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OBJECTIVES:

It is expected that at the end of the lectures that will be given on the Ecological Adaptation of the Animal Kingdom , the students would be able to comprehend the following:

The distribution of diversity of animals in the three main ecological habitats

The challenges and hazards that is inherent in each of the habitats and how animals survive through the challenges.

The different forms of adaptive features possessed by the animals in the different habitats which enable them to survive.

INTRODUCTION:

The diversity of animals in the animal kingdom are adapted to living in three ecological habitats in the biosphere. The habitats include ;

Aquatic Habitat (Water)

Terrestrial Habitat (Land)

Arboreal Habitat (Tree tops/branches)

In each of the habitats there are hazards and harsh environmental conditions which animals in them have to withstand in order to survive. Animals which survive and live successfully in the habitats become adapted to life in the habitat.

Adaptation therefore implies the possession of structural/morphological, physiological and behavioural properties which enhances the survival of an animal in its habitat or environment.

Each of the three habitats are not homogenous. They are further differentiated into sub types.

The Aquatic habitat has three sub types which include **Marine habitat**, **Brackish water or Estuarine habitat** and **Fresh water habitat**

The Terrestrial habitat comprises **Forest habitat**, **Savanna habitat** and Desert **habitat**

The Arboreal habitat comprises the tree Trunks , **Branches** and the **Leaves**

For animals to adapt in their environment, they must have the ability to face the following problems in the habitat ; food scarcity, water scarcity, air(availability of oxygen in the habitat), temperature(low and high) keeping body warm and coping with changes in weather and climate in the habitat. They must also be able to defend themselves or be able to escape from enemies (predators).

ADAPTATION CHARACTERISTICS

Animals exhibit three major types of adaptive characters which enable them to survive in their habitats. They are **Anatomical / Morphological** adaptation, **Physiological** adaptation and **Behavioural** adaptation.

Anatomical/ Morphological : This is also known as **Structural** adaptation. It concerns the body shape and appendages that help the animal survive as in getting food, moving and escaping from danger, breathing, adjusting to changes in the environment within the habitat.

Physiological: This is about the functioning of the body system so that the animal can respond to changing conditions in the habitat. E.g. temperature regulation, production of poisonous venom and secretion of slime by different animals.

Behavioural: This involves special ways animals react or behave to survive in their environment when they are faced with challenges or threat. E.g. migration, herding/flocking, vocalization, mimicking and fast running .

AMINAL ADAPTATION IN AQUATIC HABITAT:

The general characteristic of animals in aquatic habitat are;

Their body is spindle shaped and steamline e.g. seals , sea lions , sharks etc

The head is elongated into a short neck for free movement.

The external ears (pinnae) is reduced and so reduce resistance

Gills are present for respiration in water

Fishes have fins while the turtles , sea lions have paddle-like fins for their movement in water

Secondary aquatic animals like the crocodile, frog and turtle have nostrils at the apex of their head and lungs for breathing when on land.

Challenges in Marine Habitat

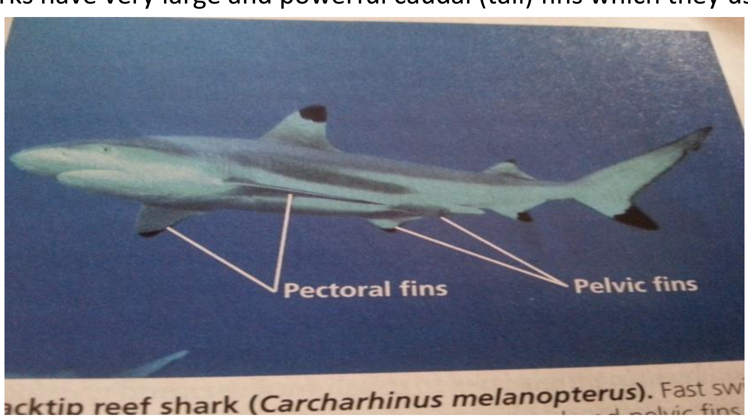
The major problems in the marine environment which animals face are **Salty water**, **Dehydration** , **Temperature changes**, and **Water pressure**.

Some marine fish prevent high concentration of salt inside their body by excreting salt with water through their gills. The sea birds drink salt water and eliminate the excess salt through the nasal gland or salt gland into the nasal cavity before it is sneezed out. Whales do not drink sea water. They obtain their water from the food (other animals) they eat.

The Turtle and Dolphin move in water with their stout and strong paddles instead of fins in order to steer their massive body sizes.

Marine Animals

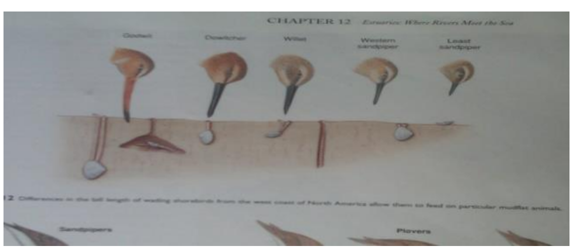
Whales and Sharks have very large and powerful caudal (tail) fins which they use to propel themselves in the sea water.

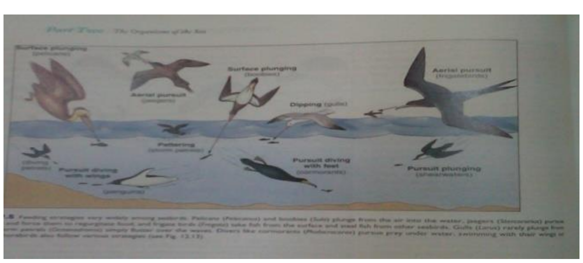


Whales can remain under water for long periods. This is possible because the lungs can exchange about 90.% of the oxygen in the air it takes in from the atmosphere and store it.

Animals in the oceans withstand cold icy condition that often occur in their habitat because they are ectothermic i.e their internal body temperature is the same as the temperature of the surrounding water. The shadowy fish (T*rematomus bernacchii)* which lives in the ice-cold water in the Antarctica region possesses an anti-freeze substance in the blood which prevents blood clotting. The marine mammals are endothermic. They have large layers of fat on the inside of their body which insulate them and prevent heat loss in cold periods. Some marine animals migrate to warm waters during the cold winter time. The whales cope with high pressure of water on them when the move deep down the sea water by collapsing their lung and rib cage as they dive down the water. The sea turtle which can dive 3,000 ft below the sea surface dose so by collapsing the lungs and flexing its shell. This helps it to withstand high water pressure above it.

The aquatic marine birds possess long legs for walking in shallow part of the water and long beaks with which they catch fish under water. Some e.g the Pelican which feed on fish have modified wings for diving under the water to capture fish.





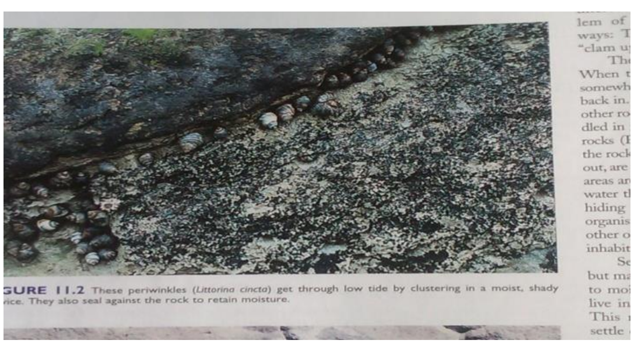
The Brackish Water/Estuarine Habitat

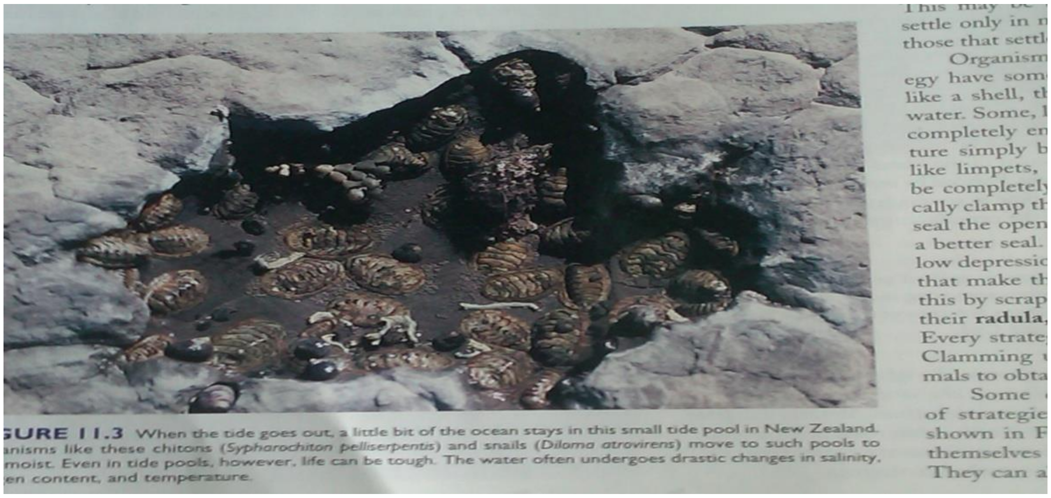
The challenges which animals face in this habitat are those of salty water which is less than that of the marine . Other challenges include ; tides, waves and desiccation. Animals in the habitat have the adaptive features for aquatic life. They possess gills; have stream line bodies and fins. The salt concentration in brackish water habitat is about 0.5 to 30 parts per thousand .

The Mudskipper (Periophthalmus) is fish like creature that lives in the estuarine environment. It has a small elongated body which reduces water resistance during movement in water. The long tail aids it in balancing as it skips with its two fin like limbs on the anterior part of the body.

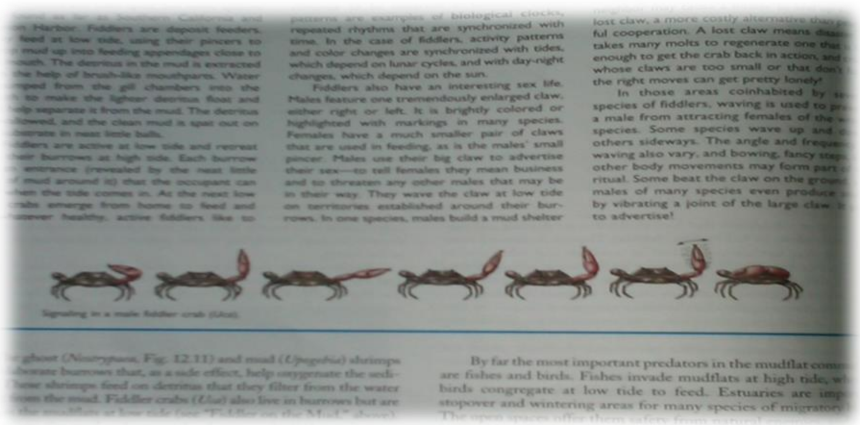
They live in burrows in high tides to avoid being washed away by tidal water as it receeds. The inner surface of the mouth is vascularised for direct intake of oxygen from the atmosphere. Also body surface is constantly moist and oxygen in air also dissolves and enters the body. They are more active during low tides .

The Periwinkles are another organism that habit brackish environment. They survive in low tides by clustering in moist shady crevices and pools of water in rocks. They have sucutorial foot for attachment to objects.





The fiddler crabs are also found in the estuarine environment. They are deposite feeders .



They feed at low tides. The large cheliate or pincers arm is used for feeding and attack/defence. With the pincers appendage it scoops mud and passes it on to the feeding smaller appendages near the mouth. The male fiddler crabs also use the pincers appendage which is large and brightly coloured during mating as an attraction for the females.

Barnacles and Bivalves which live in tidal zones face the problem of desiccation at low tides. They survive by possessing hard calcareous shells which protect them from the pounding of waves. Crabs are also protected from waves by their flat body shell.

Some organisms such as bivalves, snails and oysters enclose themselves in their shells during low tides to avoid desiccation and loss of water. Earthworms and crabs hide themselves in the mud to avoid heat.

FRESH WATER HABITAT

Challenges in the Fresh water Habitat

Osmotic problem:

The fresh water habitat has very low salt content. Its salt content is about only 1% of that of marine habitat. Because of the low salt content of the surrounding water, animals in the habitat face the problem of (**osmosis**) absorbing excess water into their body. The liquid inside their internal environment has a higher salt concentration than the surrounding fresh water. In order to avoid this problem some fresh water animals adapt by secreting the excess water in their urine.

Invertebrates in fresh water maintain osmotic balance between their internal environment and the fresh water by active up take of salt from the water.

Changes in Temperature:

Animals adapt to temperature changes in the fresh water habitat by behavioural means. In the cold season fresh water animals seek and move to warm waters or part of the water (edges) that are warmer. Some look for hiding places (niches) where the water is warm . Animals in shallow waters move up to the surface when the lower water is cooler.

Availability of oxygen:

Oxygen in water is extracted through the gills by aquatic animals. In some invertebrates e.g. amoeba it is through the body surface. When there is decrease in the concentration of oxygen some animals locate cooler portion of the water with high oxygen concentration. When there is low concentration of oxygen (Hypoxia) there is regulation of ventilation or circulation of water through the respiratory organs – Gills. E.g. Sponges and rotifers in the movement of cilia to make more water flow past the gills. The fresh water crayfish also increases the ventilation of the gills. Fishes increase the pumping out of water through the operculum and gill slits.

THE TERRESTRIAL HABITAT

The terrestrial habitat comprise of soil, forest, savanna and desert environments. Animals in the terrestrial habitat therefore show some specific adaptations in their environments. The adaptations are:

Fossorial (subterranean) Adaptation: This is found in animals that live under the soil. Their adaptive features include; Possession of organs for digging. The animals dig for food, to hide and for shelter.

Their body is cylindrical in shape – reduces resistance as the move through the soil. E.g. earthworm,


earthworm


mole cricket

The head is small and tapers anteriorly to form a burrowing snout. Neck and pinnae reduced to avoid obstruction during quick movements through holes. Some have short tails. Mostly the eyes are small and may be non-functional. The limbs are short and strong. The paws are broad and stout with long claws and extra digging structures e.g mole cricket fore limbs modified for digging.

Cursorial Adaptation.

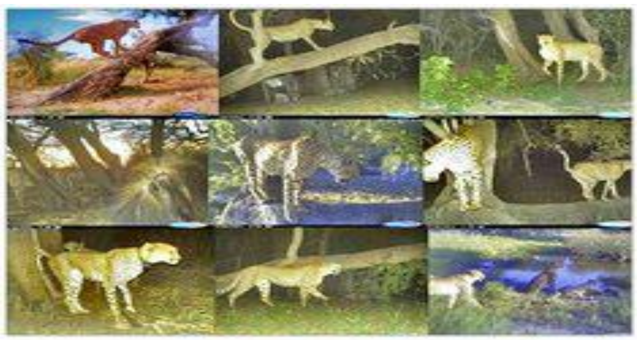
Cursorial adaptations- Involves running e.g some animals in grass land habitat (savanas) are fast running (predators) to catch prey; and also others (prey) are fast running in order to escape from the predators.

Their adaptive features include:

Reduced **Neck and stream-line body –reduces air/wind resistance.**
Bones of palm –(carpals and metacarpals) compact , may be fused as cannon bone.
Fore- arm bones - (ulna) and shank bones (fibula) reduced.
Distal segments of limbs –radius, tibia and cannon bones elongated for increase in stride.
Limb movement restricted –forward and backward.

Scansorial or Arboreal Adaptations.

Arboreal animals are animals that spend most of their time or all of it on trees or climb rocks and walls. . Examples are the flying squirrels , tree sloth , monkeys, leopard, chameleon and gecko. They are common in tropical forests where trees are thick and above 30m tall. Some of them never come down to the ground but constantly remain on tree tops jumping from branch to branch.



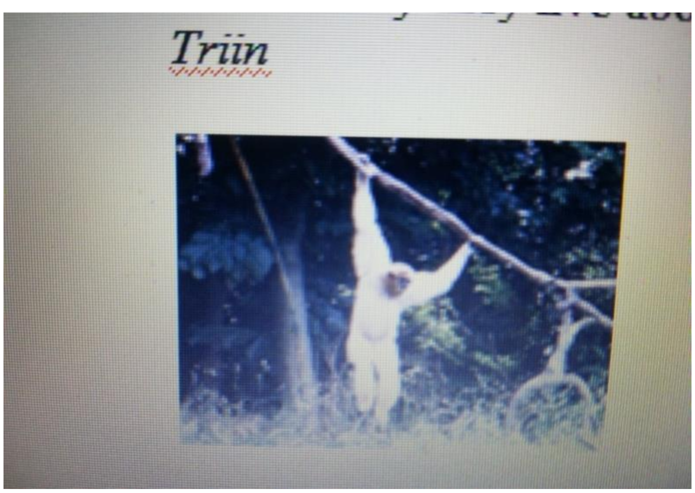
Leopard

Adaptations:

Some have small body size which helps in movement through thickets in the habitat.. They have clawed or sticky feet. Their tails are prehensile -long and stout for holding branches of trees. Some have huge claws that they use to hang their body weight as they hold branches E.g the Sloth. They have stretchy membrane between their legs or toes which allows their extensive gliding. Limbs are elongated and helps them to stretch and cross gaps between branches and trees, and also to reach out an collect food from the tip of the branches. Chests, girdles, ribs and limbs are strong and stout.
Hands and feet –prehensile (ie adapted for holding/catching- E.g Primates and Marsupials (with opposable digits).Digits 2 or 3 e.g - Chameleon. Claws elongated (squirrels). Adhesive pads at tip of digits eg tree frog (hyla).

chameleon

Other features of arboreal adaptation are Brachiating. This is fast rapid movements by primates hanging and moving from branch to branch on trees. The gibbons are very good brachiates. They have elongated limbs which enable them swing easily and grasp branches.



gibbon

The tree Anteater uses its prehensile tail as a third arm for stabilization and balance, while the claws help in grasping and climbing the branches **.**

Arboreal snails use their sticky slime and their foot in climbing trees since they have no limbs

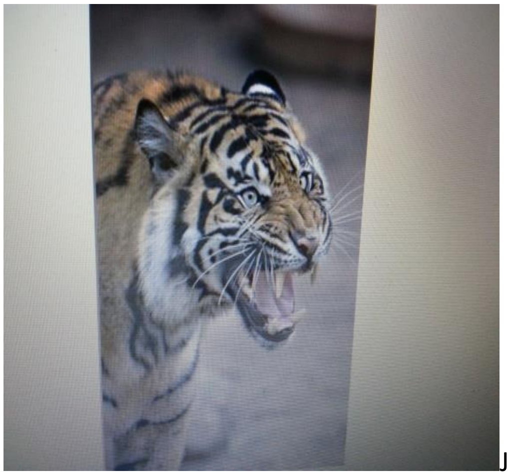
THE FOREST HABITAT

Here different variety of animals exist. The life in the forest is full of danger and struggles. The major challenges of forest animals are;

Wet environment due to constant rainfall.   Scarcity of food due to high population of animal. Lack of space because of thick covering of plants which makes fast pursuit and escape movement difficult. Insufficient   light penetration hinders vision of both prey and predators.

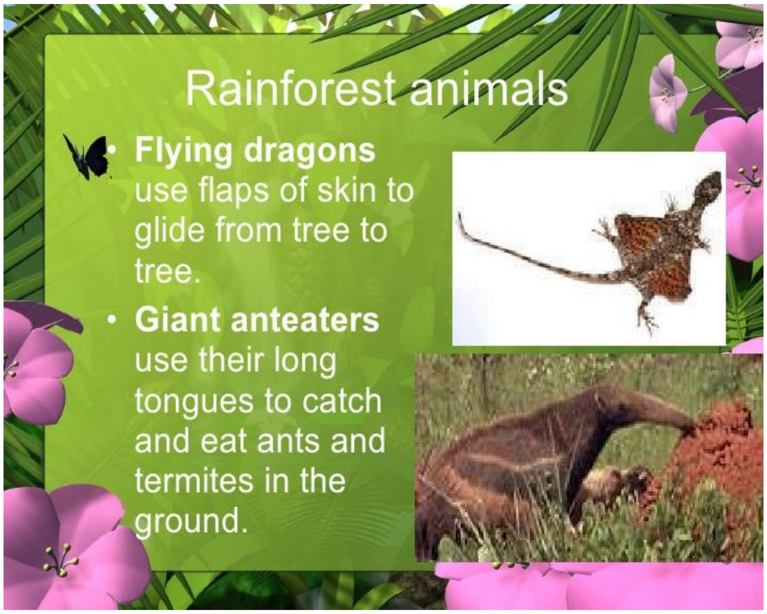
Adaptation; Animals have thick fur covering their skin against wetness by rain. E.g the Gorrila and the Chimpanzee.

The Jaguar and other large carnivores have muscular and limbs with strong claws for dragging down preys. They also have powerful jaws and strong stout teeth for feeding.



Jaguar

The toucan birds and parrots have big and strong bills which they use for breaking nuts. Spider monkey uses long tail for hanging.



Flying dragon in picture glide/fly with aid of skin flaps. Ant eater has long snout and long tongues to catch ants –see picture.

Chameleons use camouflage to hide from enemies

Some poisonous frogs like the poison dart frog warns predators of their danger by their body bright colour.

Adaptations in Wood pecker : Feeds on insects on wood bark. Has strong stout beak for peaking the wood. The beak attached to head by strong muscles. Skull bones are thick and shock absorbing muscles in between the head and beak eye lids closed – avoid particle entry / propping out eyeball. Long and strong claws for gripping on wood trunk.

Adaptation in Savanna Habitat :

Animals in the savanna face problems of water scarcity , predation , harsh weather/climate and and food. They are adapted to these problems in the following ways;

1. Migration in search of food and water,
2. Hibernation in harsh seasons.
3. Camouflage and herding for defense and escape from danger e.g zebras, elephants ,

   buffalos etc;
4. Smaller animals have colour of grass /tan fur ; burrow.
5. Most of them are Nocturnal in order to avoid the hot day time weather and predator.

   Carnivores – colour aids hiding – cheetah , fast running -65mph.
6. They have long legs , sharp sense of smell, sharp hearing and camouflage for escape from enemies.

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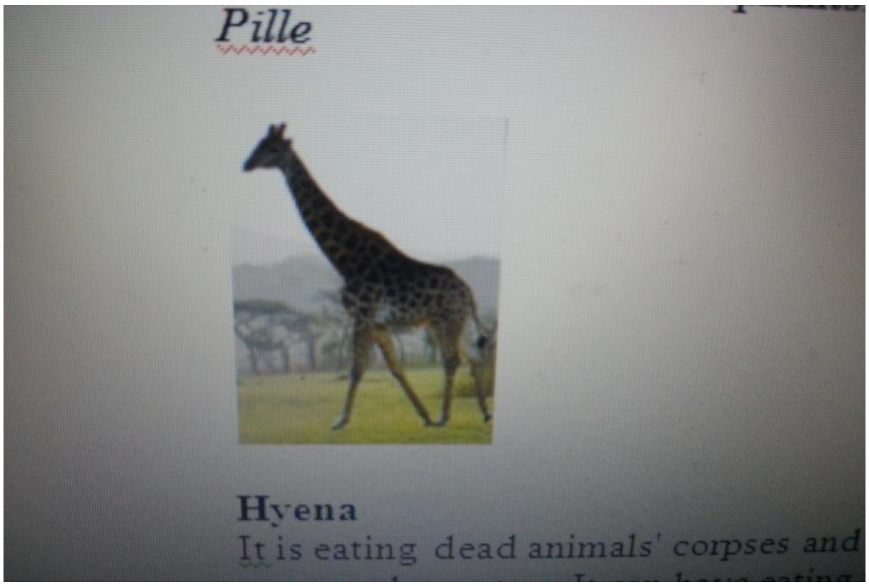
Elephants obtain water by breaking baobab tree trunk which store water.

They can sense water 3 miles (about 5 km) away.

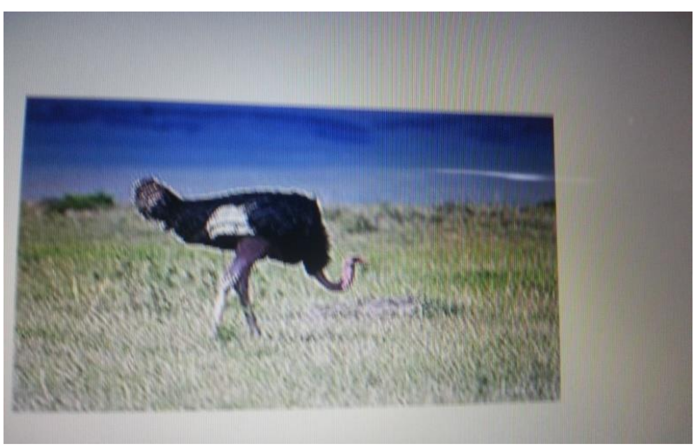
Elephants use their trunk for feeding from tall tree branches and the tusks for defence/offence.

Some savanna animals may get dormant during water scarcity period.

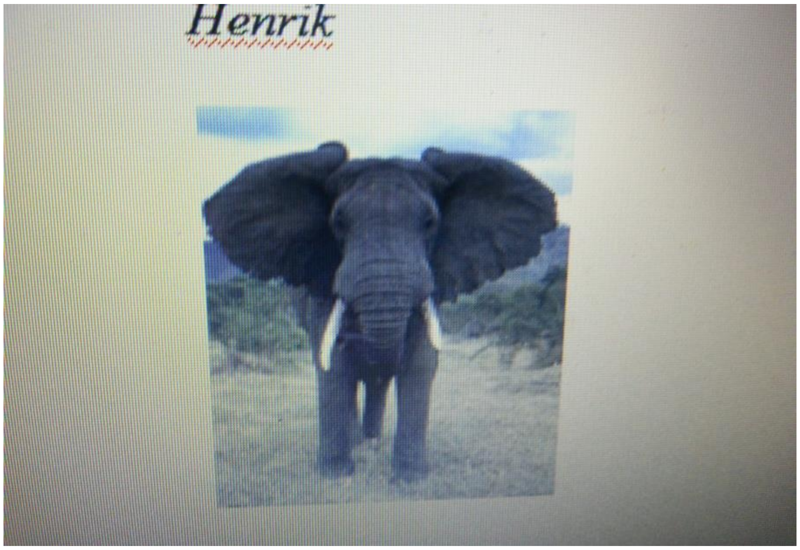
They escape from Fire Hazards by flight for winged ones like insects and birds, and by fast running in large animals eg elephants, ostrich lions while rats and snakes hide in holes.



The giraffe's long neck enables to get food from the tall savanna trees. It also helps it to view his surrounding and see the enemy from afar. The long leg helps to run with long strides away from danger.

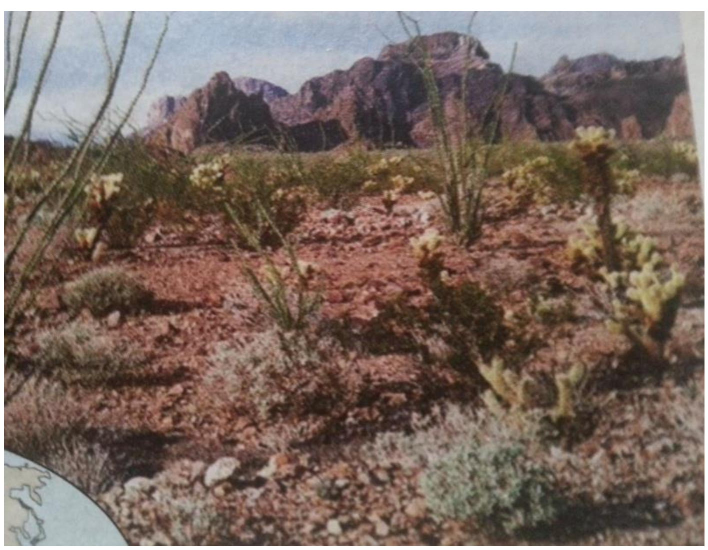


Ostrich has long strong legs which it uses for running away from danger. When it raises the long neck it is able to see an enemy from far.



The elephant uses the long trunk for reaching high tree branches to collect leaves. The strong tusks are for defence and offence.

The Desert Habitat.



The major challenges which desert animals face are water scarcity, extremes of temperature, and food scarcity. They also have to contend with the sandy wind and air in their environment.

Desert animals show two kinds of adaptations that helps them to survive in their habitat.

The adaptations are Physiological adaptations and Behavioural adaptations

Most animals in the desert obtain their water from the food they eat e.g succulent plants, fruits, and from the blood and tissue of animals they eat in the case of carnivores. Some develop hygroscopic skin which help in absorption of moisture from air e.g sand lizard and horned toad. Desert rat satisfy water need by metabolic synthesis. Animals have reduced surface areas which minimize drying.

Thick exoskeleton as present in spiders an scorpions make skin impermeable and prevent water loss.

Scorpion

The Camels store water in tissues all over the body.   Desert lizards store water in their large intestine.

Some animals in the desert have few sweat glands or they may be completely absent as a means of conserving water.

Behavioural adaptations include being inactive during the daytime and being active at night in order to avoid hight day temperature . Most animals in the desert search for food at night when it is cool.

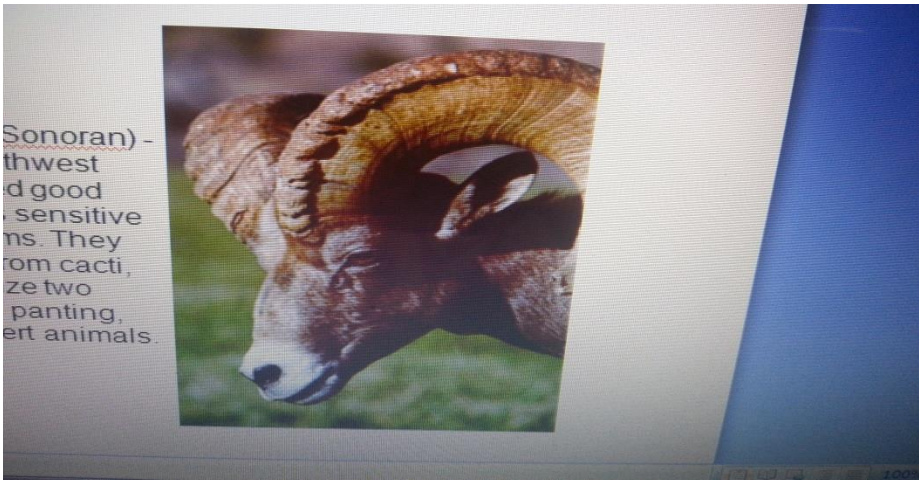
Animals in the desert store their fats in lumps instead of all over the body. This helps to reduce body heat.

A behavior of animals in the desert is seeking and staying under shades of shrubs, trees, and shadows Cast by boulders (big rocks) in order to keep away from the hot sun. Some will burrow and enter the soil for shelter. Some desert snakes bury their body in sand to avoid detection and catch a prey.

Animals in the desert form and release concentrated urine in form of uric acid as a means of conserving Water. They possess salt glands which helps in secretion of salt without loss of water.

Desert Bighorn Sheep (Mojave, Chihuahuan, Sonoran) - indigenous to the hot desert habitats of the Southwest region of the United States. They use their hooves and horns to remove spines from cacti, then eat the juicy insides

Desert Bighorns utilize two mechanisms for cooling -- perspiring, and also panting, which is a fairly uncommon adaptation for desert animals.



Desert sheep

Desert aminals produce poisonous secretions from their skin for defence.

E.g toads , Snakes and spiders release poisonous secretions and venom for defence and attack.

Desert animals protect themselves from dust and dusty air by the possession of thick eye lids.

They also have reduced ears and nostrils which are provided with valves and scales to prevent entry of dust.